

“IoT COMMUNICATION PROTOCOLS”

Course Code: BCO701



A Report submitted to

Visvesvaraya Technological University

In partial fulfillment of the requirement for the award of degree of

Bachelor of Engineering in

Computer Science and Engineering

(IoT, CyberSecurity including BlockChain Technology)

Submitted by

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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(IoT, CYBER SECURITY INCLUDING BLOCKCHAIN TECHNOLOGY)**

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No.14, Raghuvanalli, Kanakapura Main Road Bangalore- 560109

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CERTIFICATE

This is to certify that the “**IoT COMMUNICATION PROTOCOLS (BCO701)**” activity is a **bonafide work carried out by Anushree K N , 1KS23IC008**. In partial fulfillment for the award of **Bachelor of Engineering in Engineering (IoT, Cyber Security Including Blockchain Technology)** of the **Visvesvaraya Technological University, Belgaum** during the year **2026 – 2027**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The report has been approved as it satisfies the academic requirement in respect of course prescribed for the Degree.

Internal assessment marks awarded:

Marks Obtained:	
Total Marks:	100

Signature of Course In-charge
Ms Rachana V Murthy

Signature of HOD
CSE-ICB

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ABSTRACT

Arduino is an open-source electronics platform widely used for developing and prototyping embedded and IoT applications. The Arduino IDE provides an integrated environment for writing, compiling, and uploading programs to Arduino boards. This report includes the installation and basic configuration of the Arduino IDE, providing an understanding of the software environment required for Arduino-based development. Virtual Labs provide web-based learning environments that allow students to perform experiments and practical activities through digital simulations without requiring direct access to physical laboratory equipment. They provide an interactive and flexible platform to configure experiments, perform procedures, observe results, and repeat experiments safely. Virtual Labs are particularly useful in electronics, embedded systems, computer science, and IoT education, helping students understand programming, hardware components, sensor interfacing, input-output operations, and system behaviour while reducing dependency on physical laboratory infrastructure. Raspberry Pi is a compact single-board computer that provides GPIO-based interfacing capabilities for IoT and embedded applications. The report introduces the Raspberry Pi and demonstrates its practical use through experiments involving LED control and an HC-SR04 ultrasonic sensor. These experiments provide an understanding of GPIO programming, digital output control, sensor interfacing, and distance measurement, forming a foundation for developing practical IoT systems.

Keywords: *Arduino, Arduino IDE, IoT Virtual Labs, Internet of Things (IoT), Raspberry Pi, GPIO, LED, HC-SR04, Ultrasonic Sensor, Embedded Systems, Hardware Interfacing.*

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